

PREPARED BY ADIL ALVI

MTH301 MIDTERM SUBJECTIVE QUESTIONS 2019 - 2010

Q1) = scalar triple product find krna tha *2

Q2) = 1 simple equation the solve krni the wo abi yaad ni lecture no 6 sy the * 2

Q3) = derivative find krna tha *3

Q4) = Equation bnani the kud statement given thi *3

Q5) = 2nd partial derivative wala 1 question tha *5

Q6) = Chain rule wala 1 question tha u.v wala *5

Dec 17, 2019

1) Square hamy diya tha us ka third part nikalna tha

2) $\ln e^{2x} \sin y$ is ki range batani thi $x=2\pi$ $y=0$

3) ak graph wala tha

4) ak derivative wala tha

Jun 18, 2019

1) Find at least two points that lie on the parabola [2 marks]

2) Find the points where the parabola, represented by the equation $y=x^2-4x$, meets x-axis.

3) Find the angle between the vectors [3 marks]

4) A line in three dimensional space passes through the point and parallel to the vector. Write down the equation of this line in parametric and symmetric form. [3 marks]

5) Find equation of normal line to the given surface at the point

2018

1) Equation of tangent plane to the surface

2) Partial Derivative

3) Evaluate Limits

4) Triple scalar product.

Dec 22, 2016

1. Sum of three positive integers x , y and z is 9. Write an equation to minimize by square of the first two variables and cube of the third in x , y . (2 marks)

2. Write total differential (2 marks)

3. Was about finding maxima, minima or saddle point. Critical point was given. (3 marks)

4. Finding double differential

5. Was also about finding maxima, minima or saddle point. Critical point was given. (5 marks)

6. Use differential method to find some function was maximum at some point 5,-5.

Jun 29, 2015

1) Double integration 5 marks

2) Find the equation of the tangent 5 marks

3) Find the gradient of the equation 3 marks

4) Absolute extrema 2 marks

5) Evaluate the integral 3 marks

6) Graph and identify in that graphs the double integral

Dec 28, 2013

For any query feel free to ask adilalvi21@gmail.com

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1) 5 marks k thay 3 question double integral se aye

2) find critical points

3) 2 graph thay

Dec 28, 2013

Two marks question: write the equation of plane. Point and vector normal to it are given. Simple one line answer!!

Find approximate change in function using differentials. Change in x and function was given.

Find double integral of given function. The integration was without limits.

Evaluate the following double integral.

$$\iint 3 - 2x - 3y^2 \, dx \, dy$$

$$\text{Let } f(x, y, z) = yz^3 - 2x^2$$

Find the gradient of f.

Find Equation of Normal line (in parametric form) to the surface

$$f(x, y, z) = xy + 2yz - xz^2 + 10 \quad \text{at the point } (-5, 5, 1)$$

May 25, 2013

$$xy + 2yz - xz^2 + 10$$

equation of line parametric form for $(-5, 5, 1)$ 5 Marks

$$\int_0^2 \int_{-2}^4 y \cos x^2 \, dx \, dy$$

find change in limits

3 Marks

$$\int_{-1}^1 \int_0^2 (1 - 6x^2 y) \, dx \, dy$$

3 Marks

long Question

$$\int_{-1}^1 \int_{-1}^1 (4 - x^2 - y^2) \, dy \, dx$$

Dec 03, 2012

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- 1) Don't waste time on lengthy chain rules!! Two questions from integration topic
- 2) One question was of double integration whose limits are given (5 marks). That was easy but lengthy as many steps were involved.
- 3) One 5 marks question: Critical point was given as (1, 1). Determine if function has relative minimum or maximum?
- 4) Two marks question: write the equation of plane. Point and vector normal to it are given. Simple one line answer!!
- 5) Three marks question: Find approximate change in function using differentials. Change in x and function was given.
- 6) Two marks question: Find double integral of given function. The integration was without limits.

Dec 8, 2012

- Q: 21: Write down the equation of a plane passing through the point (3, -4, and 2) and perpendicular to the vector $n=4i+3j+6k$
- Q: 22: Let xyz be three positive integers, whose sum is 18 and the product of two integers with square of third integers will be maximum. Write the function to be maximized in the function x and y .
- Q: 23: Find the critical point for the function $f(x, y) = x^2 + y^2$ along the line $y + 2 - 2x$ at which the absolute extrema of the function can exist.
- Q: 24: Find the equation of normal line (symmetric form) to the surface $f(x, y) = x^2 + y^2 + z^2 - 6$ at the point $(-1, 2, 1)$.
- Q: 25: Using second partial derivative test show that the function $f(x, y) = xy(24 - x - y)$ is maximum at $x=8$ and $y=8$.
- Q: 26: Find the gradient of the $f(x, y, z) = yz^3 - 2x^2$.

Nov 26, 2011

- 1) Suppose that the surface $f(x, y, z)$ has continuous partial derivatives at the point (a, b, c) . Write down the equation of tangent plane at this point. 2 Number
- 2) Evaluate the following double integral. 2 Number
[[(12xy² - 8x³) dy dx] [= integral sign
- 3) Evaluate the following double integral. 3 Number
[[(3 + 2x - 3y²) dx dy
- 4) Let $f(x, y, z) = xy^2e^x$. Find gradient of f 3 Number
- 5) Find all the critical points of the function. $F(x, y) = 4xy - x^3 - 2y^2$. 5 Number
- 6) Double integral in rectangular co-ordinates to couple area of region bounded by the Curves $y = x^2$ and $y = x + 2$

Dec 10, 2010

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